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Q1. What are the advantages of Bluetooth over RF transmission? How to implement a Bluetooth device?

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Hubli

A1. Normally, peripherals like keyboard, printer, mouse and monitor are connected to a PC using cables. This arrangement is quite cumbersome.

A solution has come in the form of Bluetooth—an inexpensive, low-power, short-range radio-based technology that connects all these devices together without requiring cables.

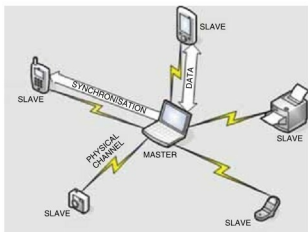


Fig. 1: Bluetooth network

Conceived initially by Ericsson, Bluetooth is a standard, small radio chip plugged into computers, printers, mobile phones, etc. The information that was normally carried by the cable is transmitted by the Bluetooth chip at a special frequency to a Bluetooth-compatible receiver. The arrangement is shown in Fig. 1.

The developers soon realised that a lot more was possible. If you can transmit information between a computer and a printer, why not transmit data from a mobile phone to a printer, or from a printer to a printer?

However, Bluetooth is not a wireless networking solution like AirPort. It is an alternative to the IrDA (Infrared Data Association) standard. Although the IrDA standard too supports wireless communication between peripheral

als and computers, it has two limiting requirements: First, IrDA devices must be very close—not more than 1 metre apart. Second, the communicating devices must have a direct line-of-sight to each other.

Bluetooth communication overcomes these requirements as it relies on radio waves. Devices can communicate within a range of up to 10 metres without needing to be in direct sight of each other. This makes Bluetooth communication much more flexible and robust. However, Bluetooth excels at low-bandwidth data transfer, and is not intended as a replacement

for high-bandwidth cabled peripherals such as external hard drives or video cameras, for which cables are still the best option.

Bluetooth headphones are another range of devices based on Bluetooth technology. These outdoor wireless headphones based on radio frequency transmission in terms of ease, convenience and mobility.

Also, a wide range of devices can be used with them, including mobile phones, media players, personal digital assistants, computers and game boxes.

RF transmission, with which we are more familiar, is also wireless. The various ranges of frequency used in the transmission come under the radio frequency spectrum spanning from 30 Hz to 300 GHz; typically, 10 kHz to 100 GHz for long-wave, short-wave and microwave radio transmission.

RF transmission can be categorised as follows depending on the medium through which the RF signals propagate:

1. Free space propagation: Signals travel in free space. Signals travelling to and from satellites use this type of radio propagation.
2. Ground wave propagation: